Multi-Physics Simulation (Application under MATLAB & Simulink)

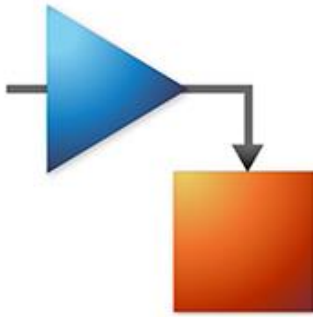
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Simulink



Transfer Function

Transfer Function - Reminder of the Principle

❑ **The differential equation of a physical system:**

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \frac{d^m u(t)}{dt^m} + b_{m-1} \frac{d^{m-1} u(t)}{dt^{m-1}} + \cdots$$

where:

- $y(t)$ is the **system output**,
- $u(t)$ is the **system input**,
- a_i and b_i are the **system coefficients**.

❑ **Application of the Laplace transform on the terms of the equation:**

$$\mathcal{L}[y(t)] = Y(s), \quad \mathcal{L}[u(t)] = U(s).$$

❑ **The transfer function $H(s)$ is defined as the ratio of the output $Y(s)$ to the input $U(s)$:**

$$H(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0}.$$

➔ **The transfer function is therefore a ratio of two polynomials in s, involving only algebraic operations: +-*/**

Transfer Function - Reminder of the Principle

- **Example of a classical differential equation for the mass-spring-damper system**

$$m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx = f(t).$$

- **Application of the Laplace transform on the terms of the equation:**

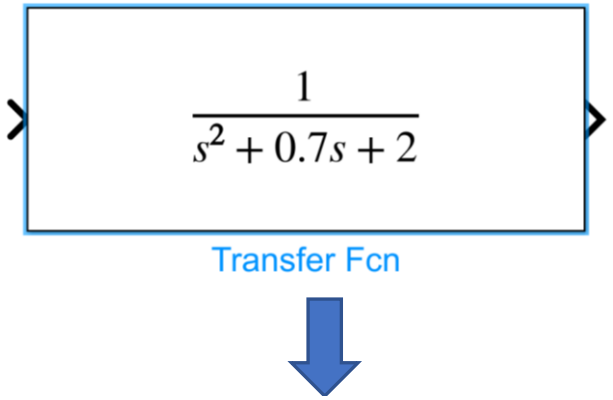
$$ms^2X(s) + csX(s) + kX(s) = F(s),$$

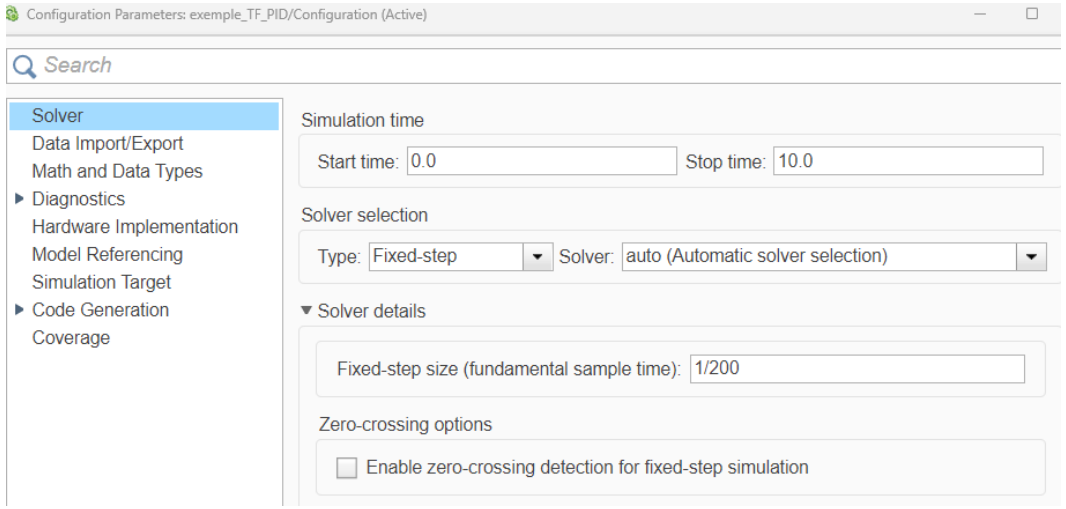
- **The transfer function $H(s)$ is defined as the ratio of the output to the input**

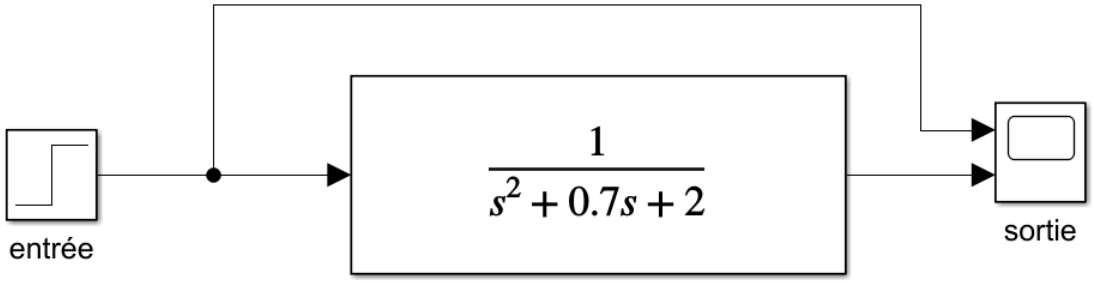
$$H(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k}.$$

Transfer Function

Tip: You can move the entire model by clicking the mouse wheel

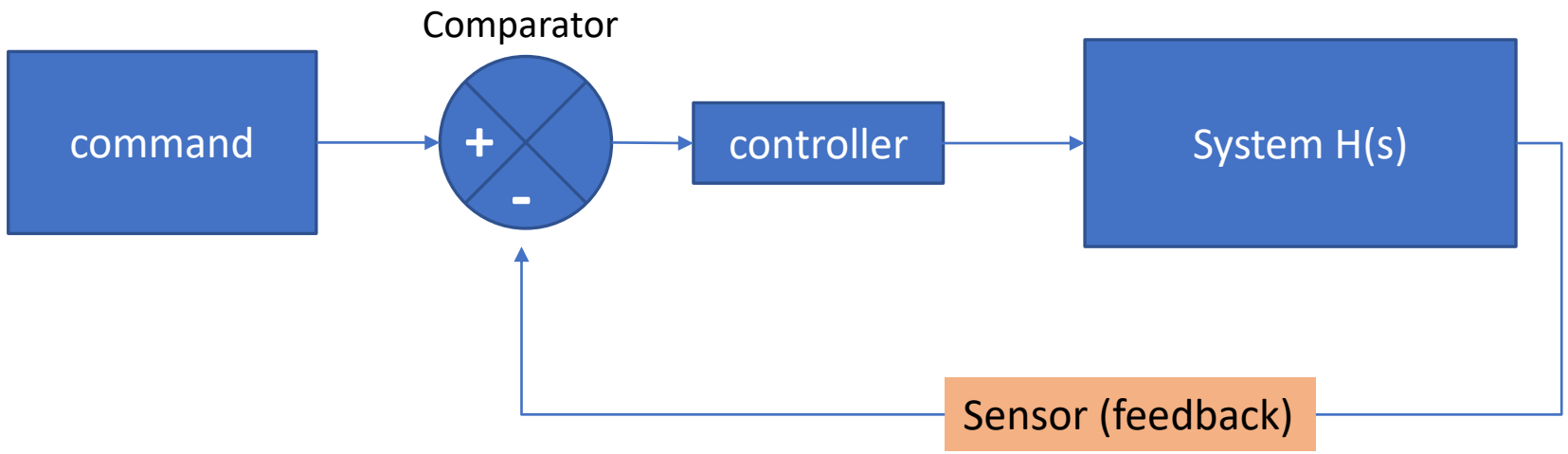




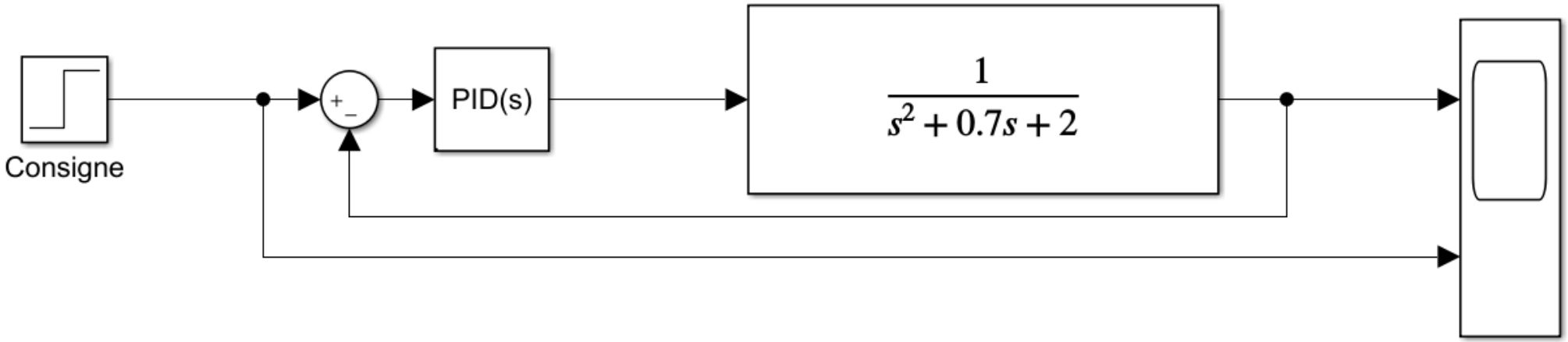


Transfer Function + PID Controller

Ajout d'une boucle d'asservissement (contrôle) pour améliorer la réponse du système

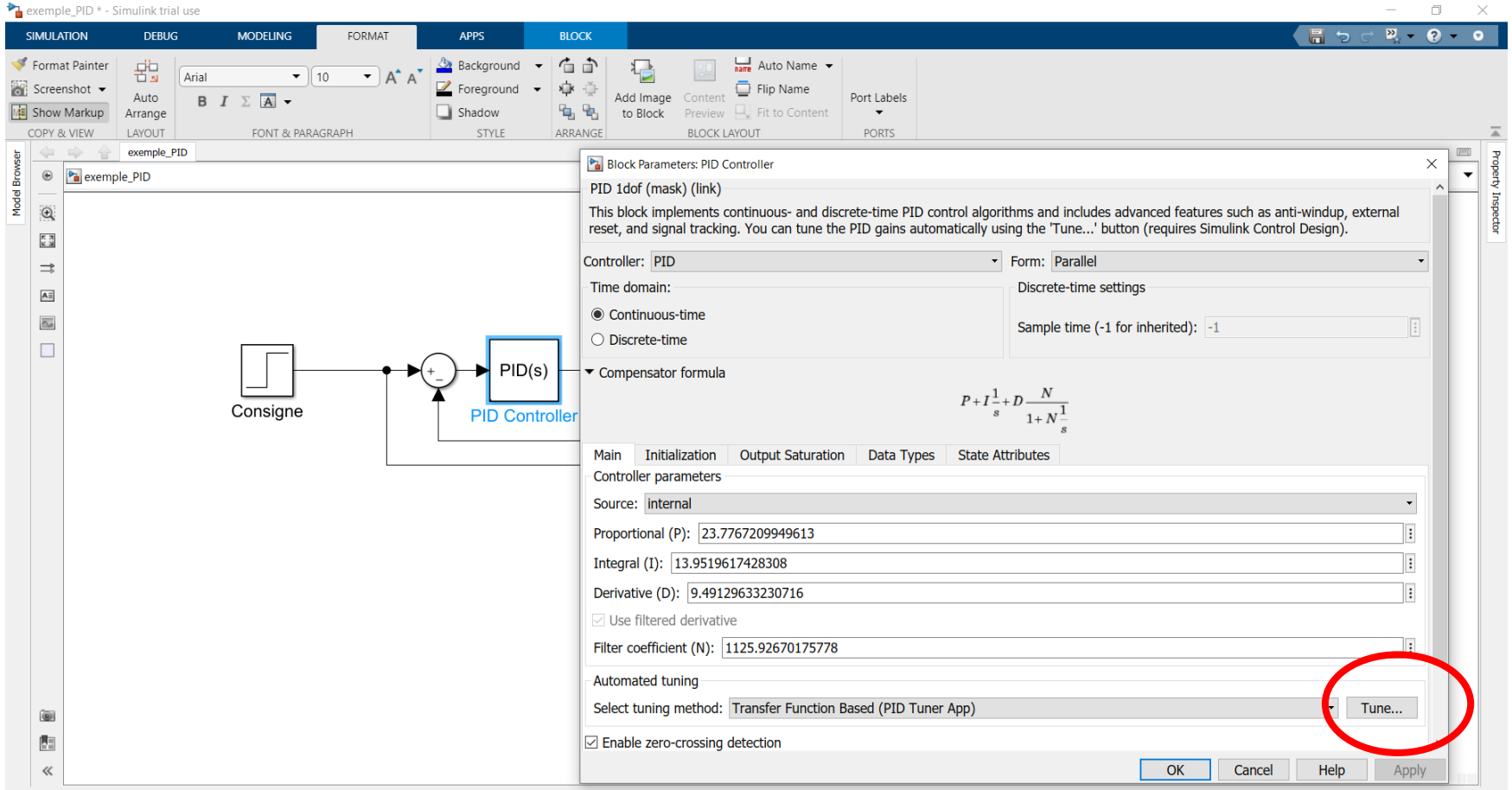


Transfer Function + PID Controller



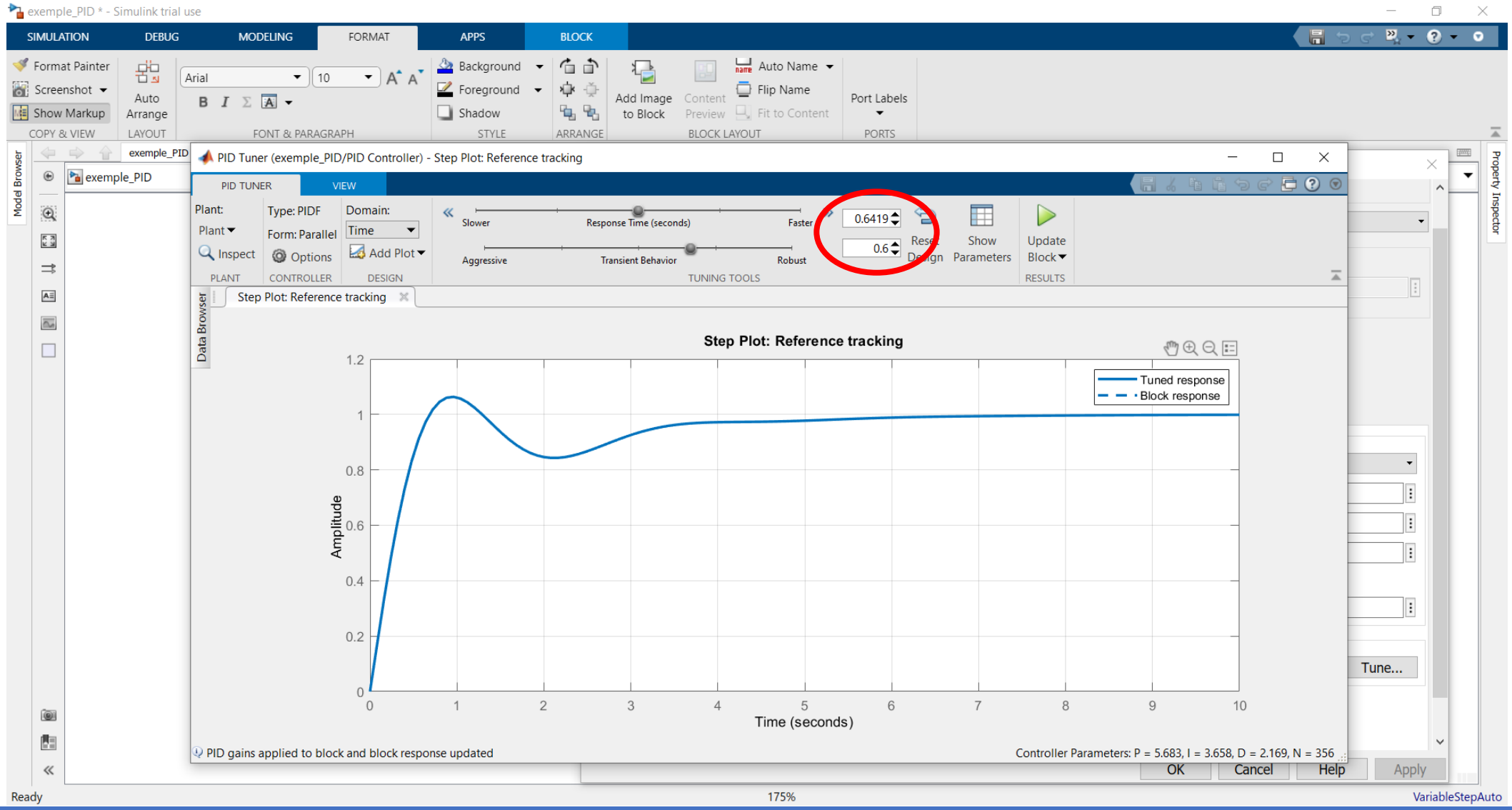
Transfer Function + PID Controller

PID tuning to meet a requirement



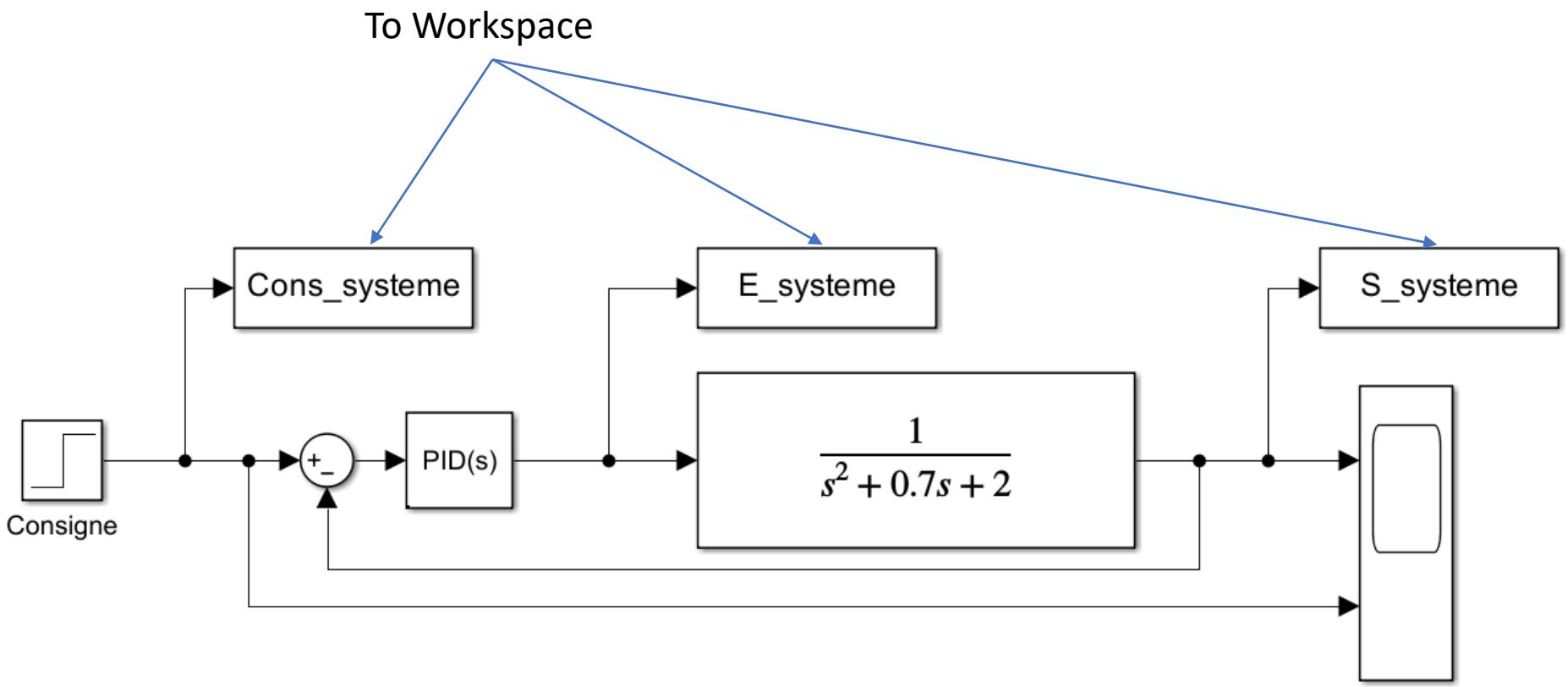
Transfer Function + PID Controller

PID tuning to meet a requirement



Save to MATLAB Workspace

Simulink variables to MATLAB workspace



Save to MATLAB Workspace

Simulink variables to MATLAB workspace

To disable the "out" structure.

